



# **Attune**

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# About Attune

## Problem addressed:

Do none of your friends like the same music as you?

Finding people who share your music taste can be super hard - especially in college, where you're surrounded by potential friends, but have no way of knowing who is into the same artists as you.

## How Attune Helps:

Attune is an iOS & macOS only app that syncs your Spotify listening history, and connects you with other users within 20 miles of you who share your taste.

Browse people's top artists and tracks, send a friend request, and turn a shared favorite artist into an actual friend or concert buddy!

# Disclaimer!

Attune prioritizes your safety and privacy.

Attune does not share your specific location - ever.

You only appear as part of a “nearby users” list for people within 20 miles.

There is no way to contact an Attune user *unless* they choose to add their own contact information (like their instagram) to their bio.

By default, only your username, top artists & tracks, and profile picture are visible.

This anonymity is intentional - your location is sensitive data, and we designed Attune with that in mind.

# Utility, Usability, and Accessibility

## Utility:

4 SUS testers' Feedback confirmed:

- Nearby users feature, and the profile browsing features, addressed a legitimate social need
- Safety concerns with location were addressed through anonymous-by-default profiles.

## Usability:

Through user testing, we discovered:

The free Spotify API tier we used limits how many accounts can authenticate with the app :(

We had to manually allow each tester on the developer side, which hurt our Usability experience.

IRL, we would apply for Spotify's extended quota mode, or upgrade our API access to improve usability.

# Utility, Usability, and Accessibility (cont.)

## **Accessibility:**

We tested with Mac VoiceOver and the macOS Accessibility Inspector. This revealed several areas of our app that were invisible or poorly described by screen readers, which we addressed using semantic wrappers.

We also audited and adjusted our color schemes, text and button size, and navigability to meet accessibility standards.

After these changes, the app was navigable via VoiceOver without visual reference, and more accessible via larger text and tap targets.

# Design & Data Structures - Models Design

The model uses `toMap()` and `fromMap()` to write to and read from Firestore, which let us read and update our user data throughout the app.

There is no separate model for artists or tracks, which are stored directly on the user document as lists.

Simple data structures helped us have more bandwidth to successfully interact with Firebase as well as two APIs.

# Design & Data Structures - Models Design

Our primary data structure is `UserModel`, which represents a user in Firestore.

## **UserModel stores:**

- a unique user ID
- username & password
- email
- Spotify ID
- top artists & tracks (from the Spotify API)
- latitude and longitude coords (from the geolocation API)
- list of friends

The model uses `toMap()` and `fromMap()` to write to and read from Firestore, which let us read and update our user data throughout the app.

No separate model for artists/tracks:

- stored directly on the user document as lists.
- Simple data structures helped with other, more complex areas (like Firebase & two APIs)

# Applied Course Topics

We applied:

- 1) GPS sensor access via the geolocator package (shoutout Food Finder)
- 2) Spotify and Geolocator API queries
- 3) Custom canvas drawing for our mascot (from title slide)
- 4) Firestore data persistence
- 5) Internationalization (English & Spanish)
- 6) Accessibility & Robustness through inclusive design choice (simple UI, sharp color contrast, big buttons & text size, voiceover-friendliness)

# CHALLENGES!!

- **FIREBASE!**
  - Small amount of prior experience
  - Big amount of debugging/inconsistency with data persistence
  - Mystery bugs
- **Spotify API free tier restrictions**
  - Unable to have as many users as we want
  - Upgrade costs money/no easy bypass
- **Platform Configuration struggles**
  - Sorry to everyone not using a mac or iPhone :(
  - We definitely tried
  - Hopefully will get resolved in the future!

**The end**